Stylized Procedural Sky - manual

A procedural sky for Godot 4. EmacEArt, Maciej. Package version 1.1. Engine: Godot 4.7 or newer, Forward+ renderer (it also runs on Mobile; Compatibility has not been tested).

1. What this package is

One shader draws the entire sky above your scene: the gradient from the ground up to the zenith, the sun disc with its glow, the haze along the horizon, and clouds drifting overhead.

There is not a single sky texture, cubemap or HDRI photo in the package. The sky is built from the settings you have at hand in the inspector. That gives you two practical differences from a sky built out of an image:

With an image-based sky you had to open a paint program and repaint the texture.

nothing running.

- **Control.** Zenith color, sun height or cloud coverage are sliders in the inspector.
- **Preview.** Every change shows up straight away in the Godot editor viewport, with

The sky is not only a backdrop. Godot can light the scene with the colors of the sky: set ambient light and reflections to Sky in the Environment, and objects standing in shadow take on the zenith color while glossy surfaces pick up the sun and the clouds. Pushing the sky toward a warm dusk then warms up the models as well, so a single setting changes the time of day across the whole scene.

Open the demo scene and the picture is already there. Nothing is generated at startup and nothing has to be run.

2. What is inside, in short

transition between them.

follow your directional light on its own, so the sky, the shadows and the scene lighting agree without extra work.

example an afterglow in the west.

makes distant ground fade and push away.

1 to 8 flat steps (a hand-drawn look) and their own drift across the sky.

lift / gamma / gain, which tint shadows, midtones and highlights separately.

- **Sky gradient** - separate colors for zenith, horizon and ground, with an adjustable
- **Sun** - a disc plus a glow corona, each with its own color and size. The sun can
- **Side gradient** - an extra color coming in from a chosen compass direction, for
- **Horizon haze** - the bright band above the ground, the same effect that in nature
- **Clouds** - five patterns, from torn wisps to solid heaps, with posterization into
- **Color grading built into the shader** - exposure, contrast, saturation, plus

38 parameters in six inspector groups. Two ready-made materials. A demo scene with a slider for everything.

3. Putting the sky into your own scene

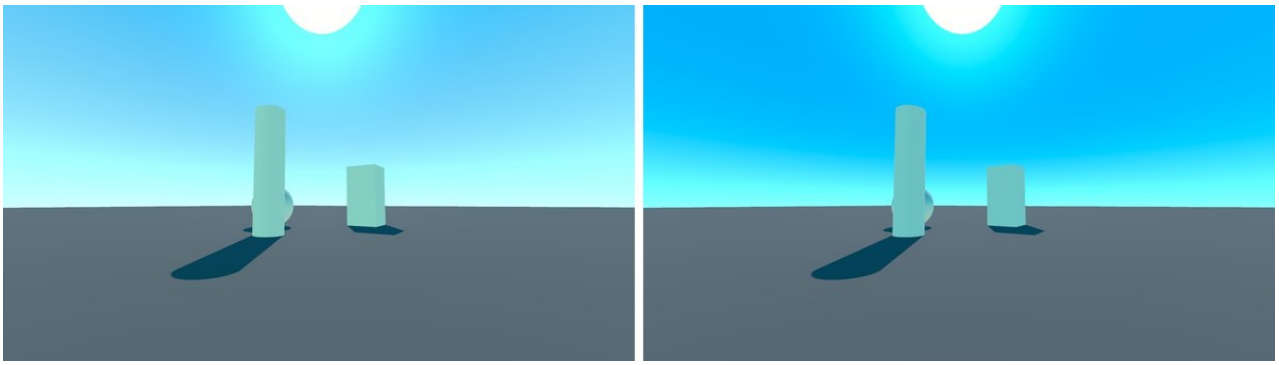
1. Add a **WorldEnvironment** node to your scene. 2. In its Environment resource set **Background** → **Mode: Sky**. 3. **Environment** → **Sky** → **New Sky**, then in **Sky** → **Sky Material** load one of the files from the Materials/ folder: EA_Skybox_Default.tres (clear daylight) or EA_Skybox_WarmDusk.tres (warm dusk). 4. Set **Environment** → **Ambient Light** → **Source: Sky** and **Reflected Light** → **Source: Sky** so the scene picks up the colors of the sky. 5. Add a **DirectionalLight3D** as the sun. Leave its **Sky Mode** at Light and Sky.

Every setting lives on the sky material. Select it in the inspector and you get six collapsible groups: Zenith and Horizon, Sun, Sky Gradient, Atmosphere, Color Grading, Clouds. The chapters below go through them in that order.

> **Before you start moving sliders:** open Scenes/SkyboxDemo.tscn and run the scene. > You get a slider panel over the picture and can watch what every setting does live, > before you type anything into your own project.

4. The settings, one by one

4.1 Zenith and Horizon - the base of the whole sky



Left: horizon_sharpness 1.0, the horizon color reaches high and blends with the zenith across most of the sky. Right: horizon_sharpness 14.0, the transition is squeezed into a narrow band above the ground with clean zenith color above it.

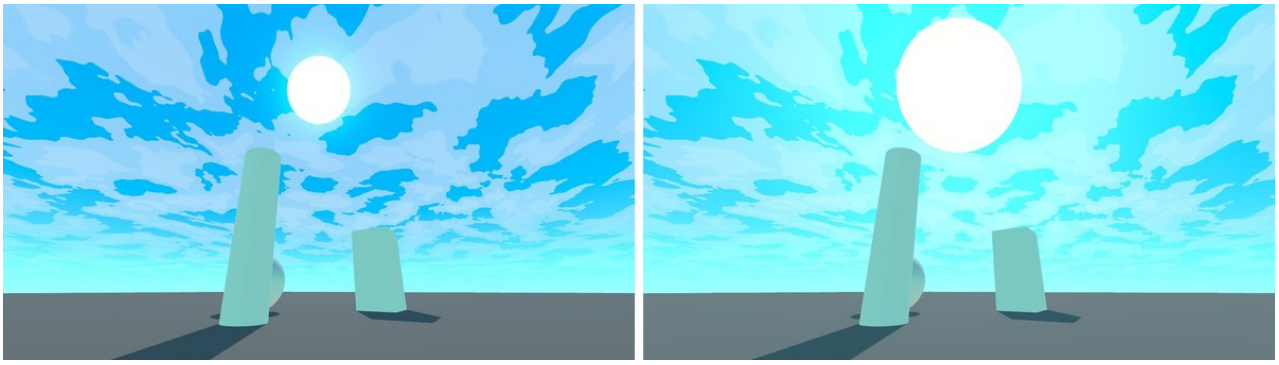
Three colors and two sliders that set up the whole picture. Everything else is layered on top of this.

Setting	Range (default)	What it does
zenith_color	color (dark navy)	The color straight overhead.
horizon_color	color (light blue)	The color at the horizon line.
ground_color	color (dark brown)	The color of the lower half of the sky sphere, which is what you see when the camera looks below the horizon.

Setting	Range (default)	What it does
horizon_sharpness	0.5 - 16.0 (4.0)	How quickly the horizon color turns into the zenith color. Low values blend across the whole sky. High values give a tight band above the ground and a clean zenith above it.
horizon_offset	-0.5 - 0.5 (0.0)	Raises or lowers the sky gradient without moving the line where the ground begins. Positive values pull the horizon color down, negative values push it up.

What it is for. This is where you set the time of day and the mood. A dark zenith with a bright, narrow horizon gives clear midday; a bright zenith with a wide warm band above the ground gives late afternoon.

4.2 Sun - disc and glow



Left: a small disc with a tight corona (sun_size 0.008, sun_glow_size 0.18, sun_glow_falloff 16). Right: a large disc with the glow spread over half the sky (sun_size 0.03, sun_glow_size 0.85, sun_glow_falloff 2.5).

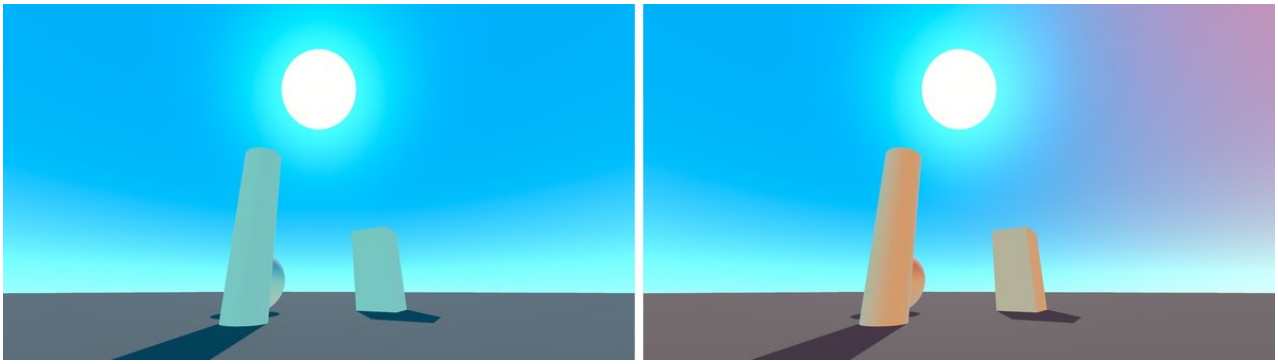
The sun is made of two parts: a **disc** with a hard edge and a **corona**, the soft glow around it. Each has its own color, so you can have a white disc inside an orange glow.

Setting	Range (default)	What it does
sun_follows_light	on / off (on)	**On** - the disc is drawn exactly where your DirectionalLight3D points from, so the sky, the shadows and the lighting stay in sync by themselves. Recommended. **Off** - the sun sits where sun_direction_manual points, ignoring the lights in the scene.

Setting	Range (default)	What it does
sun_direction_manual	XYZ vector	The sun direction used when the option above is off. Both shipped materials use this mode.
sun_color	color (white)	The color of the disc itself.
sun_glow_color	color (white)	The color of the glow. The glow brightens the sky beneath it, so bright colors have more effect than dark ones.
sun_size	0.0005 - 0.05 (0.003)	The size of the disc. The real sun seen from Earth sits near the low end of the range; higher values give a large, storybook sun.
sun_glow_size	0.01 - 1.0 (0.132)	How far the glow reaches from the disc. At 1.0 it spills across half the sky.
sun_glow_falloff	1.0 - 16.0 (16.0)	How quickly the glow fades away from the disc. High values give a tight halo hugging the disc. Low values give a gentle brightening spread wide.

□□ With sun_follows_light on and **no** DirectionalLight3D in the scene whose Sky Mode includes the sky, the shader falls back to the direction in sun_direction_manual.

4.3 Sky Gradient - a color coming in from one side

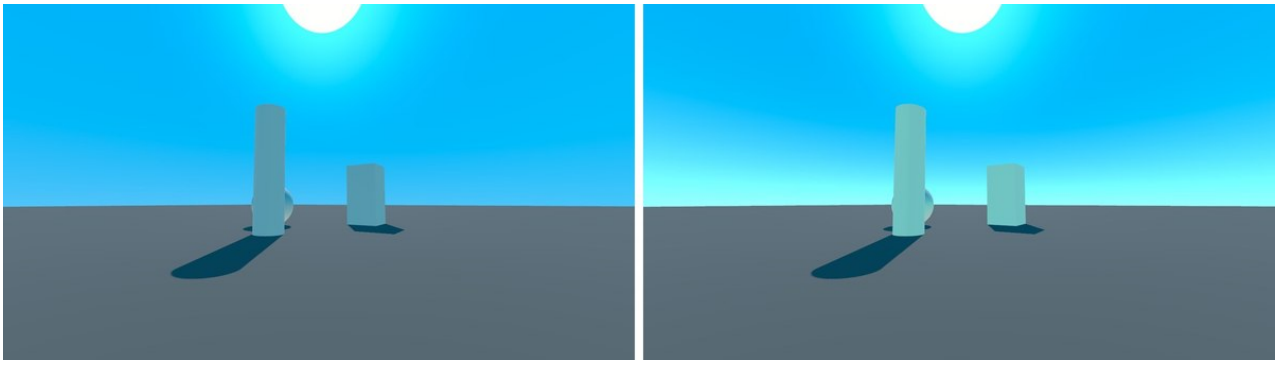


Left: sky_gradient_strength 0, the setting does nothing. Right: strength 0.9, an orange color entering from 160 degrees and warming one side of the sky.

The ground-to-zenith gradient runs bottom to top. This one runs **sideways**: it adds a chosen color on one side of the world and fades it out on the opposite side. That is how you build an afterglow in the west, a cold side of the sky opposite the sun, or an aurora over one end of the map.

Setting	Range (default)	What it does
sky_gradient_strength	0.0 - 1.0 (**0.0**)	The strength of the effect. **Zero by default, meaning off** - until you raise it, the other three settings have no effect on the picture.
sky_gradient_color	color (orange)	The color added from the chosen side.
sky_gradient_angle	0 - 360 (0)	The compass direction the color comes from, in degrees.
sky_gradient_spread	0.01 - 1.0 (0.5)	How wide a slice of sky it covers. Values near 1.0 give a soft transition across the whole dome. Values near 0.1 keep the color on one side and cut it off sharply in the middle.

4.4 Atmosphere - haze along the horizon



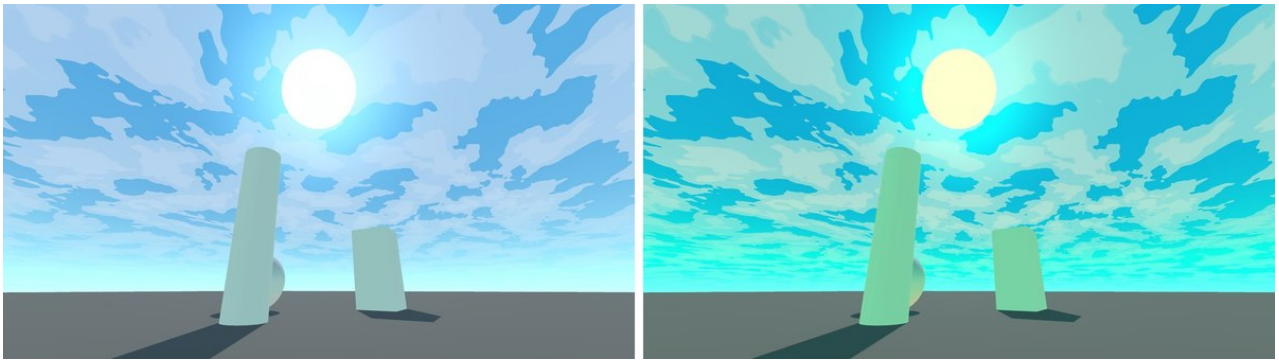
Left: atmosphere_strength 0, the sky meets the ground along a hard line. Right: atmosphere_strength 1.4, a bright band of haze lifts the horizon and pushes it away.

In nature, the further you look the more air is in the way, so distant mountains pale and turn blue. This group adds exactly that bright band along the horizon line. Without it the sky meets the ground at a hard edge and the scene looks smaller than it is.

Setting	Range (default)	What it does
atmosphere_color	color (cool blue)	The color of the haze. The haze brightens the sky, so a dark color gives a weak effect and a bright one a strong effect.
atmosphere_strength	0.0 - 1.5 (0.55)	How much haze. Zero turns the effect off.

Setting	Range (default)	What it does
atmosphere_falloff	1.0 - 10.0 (2.5)	How high it reaches. Low values let the haze climb far up the sky. High values keep a narrow strip hugging the horizon.

4.5 Color Grading - color correction with no extra nodes



Left: neutral settings (exposure 1.0, contrast 1.0, saturation 1.0, no tinting). Right: exposure 1.25, contrast 1.45, saturation 1.7, warm gain and cool lift.

This whole group affects **the sky only**. It is the final polish: instead of going back to the zenith and horizon colors and picking them again, you push the finished picture. Nothing here touches the models in the scene, so you can lift the sky without moving anything else.

Setting	Range (default)	What it does
exposure	0.1 - 4.0 (1.0)	Overall brightness of the sky.
contrast	0.5 - 2.0 (1.05)	Pushes bright and dark areas apart. Above 1.0 the picture hardens, below 1.0 it flattens.
saturation	0.0 - 2.5 (1.1)	Color intensity. Zero gives a greyscale sky, values above 1.0 give poster colors.
lift	color (black)	Tints the **shadows**, the darkest parts of the sky. Black by default, meaning no change.
gamma_midtones	color (**mid grey**)	Tints the **midtones**. The neutral point is mid grey in the color picker, which does exactly nothing. Lighter opens the midtones up, darker pulls them down.

Setting	Range (default)	What it does
gain	color (white)	Tints the **highlights**, the brightest parts. White by default, meaning no change.

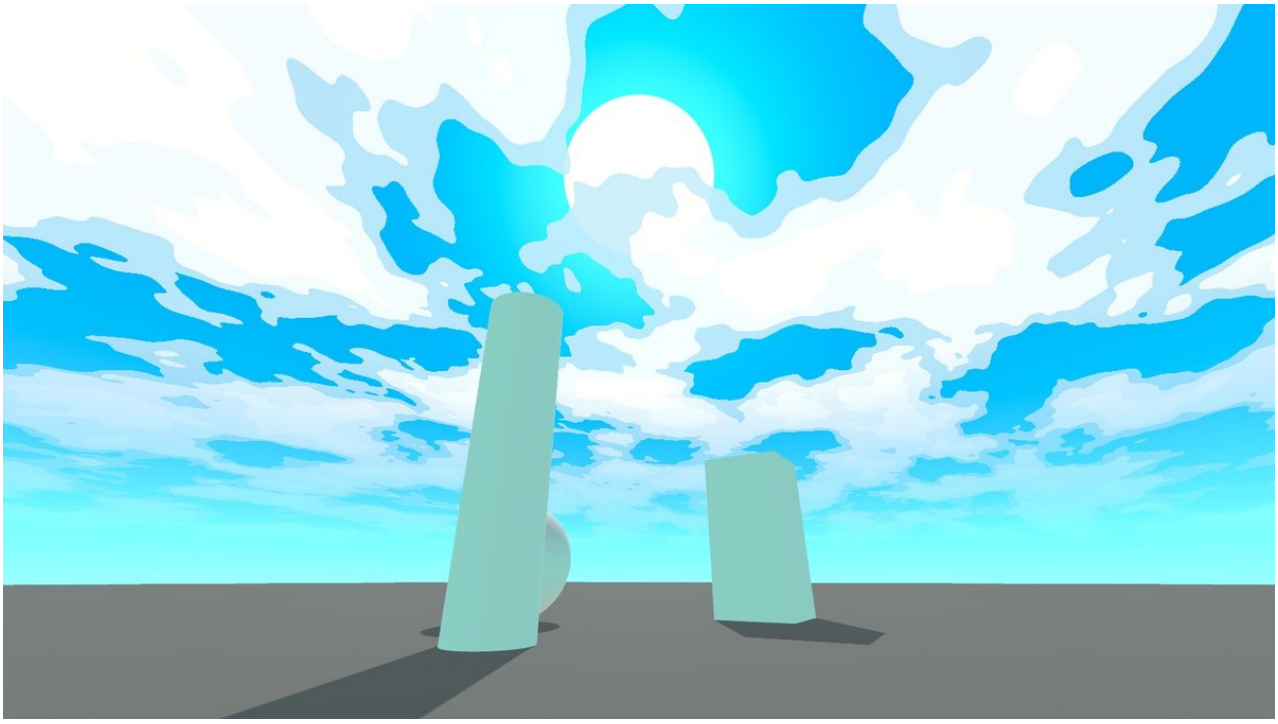
A typical use of lift / gamma / gain: a cool lift with a warm gain gives the color split most sunset photography stands on, with the shadows going blue and the highlights going orange.

4.6 Clouds

 Clouds are off by default. Tick clouds_enabled or nothing in this group will do anything.

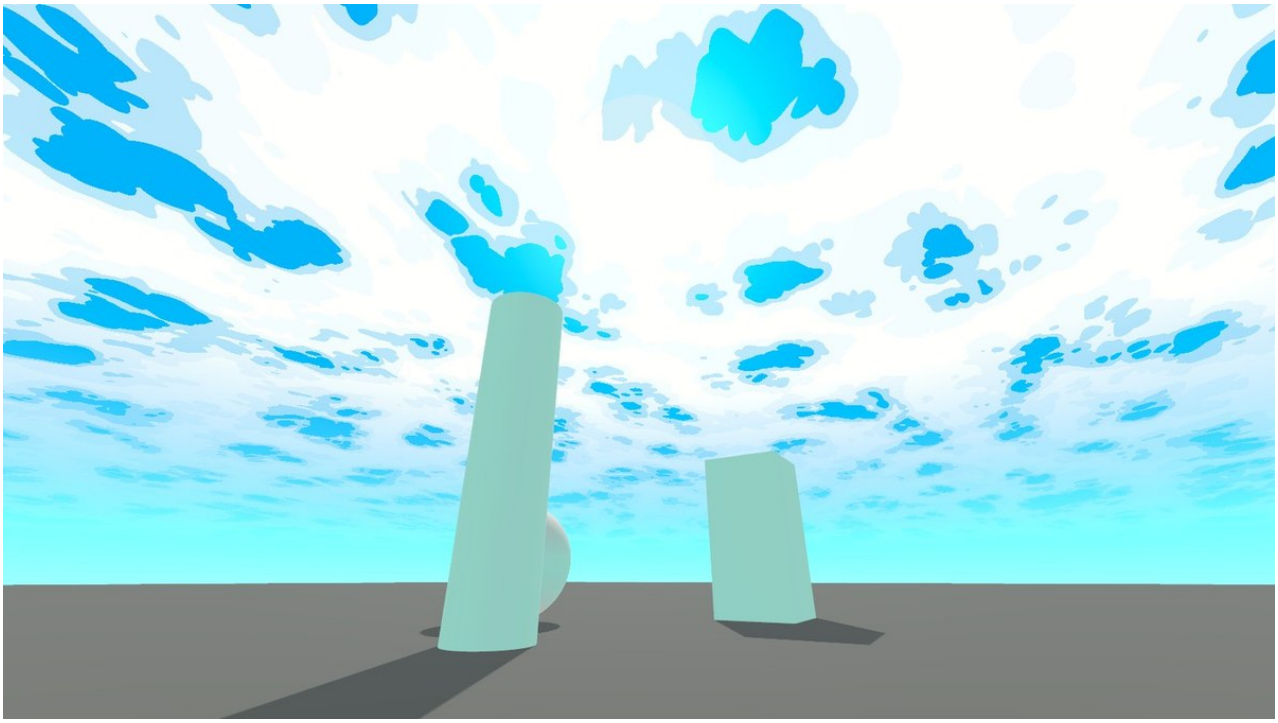
Five cloud patterns

cloud_style picks the pattern. Values run from 0 to 4; the control is a slider, but only whole numbers count.



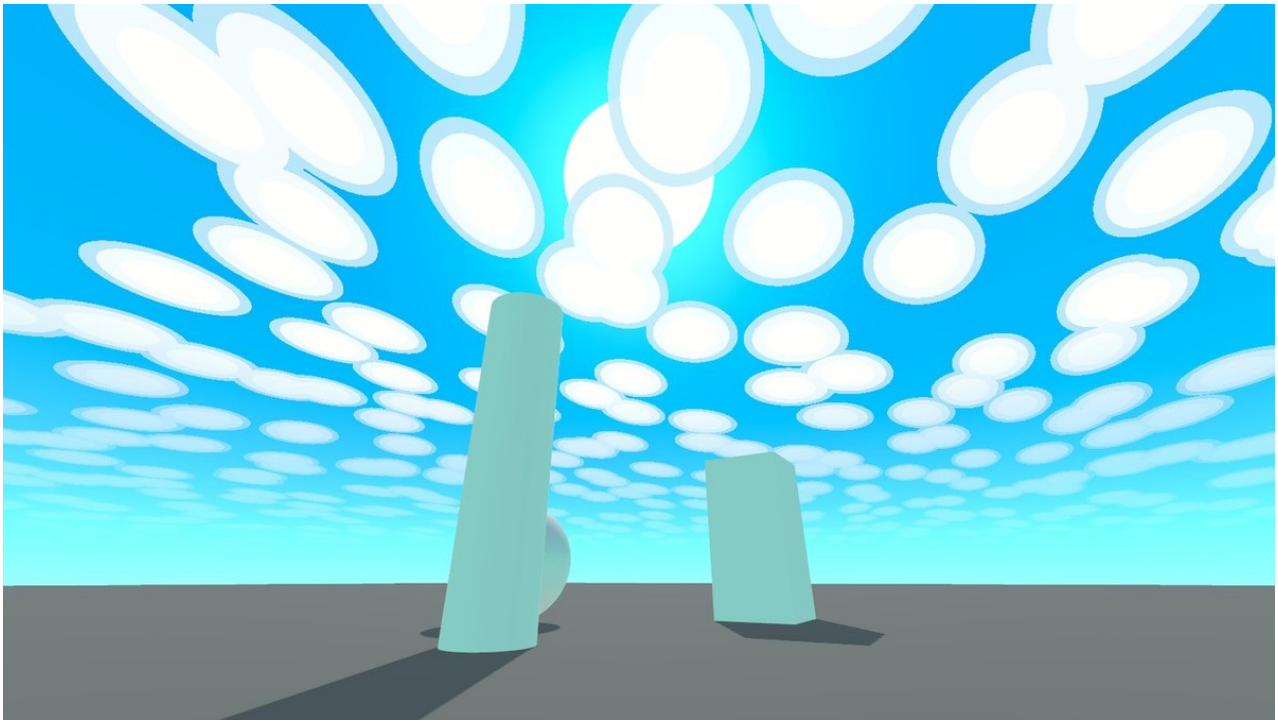
cloud_style 0, Standard: even puffy patches, a classic fair-weather sky.

0 - Standard. Puffy patches of mixed sizes with lightly torn edges. The all-purpose pattern for ordinary clouds over an ordinary scene.



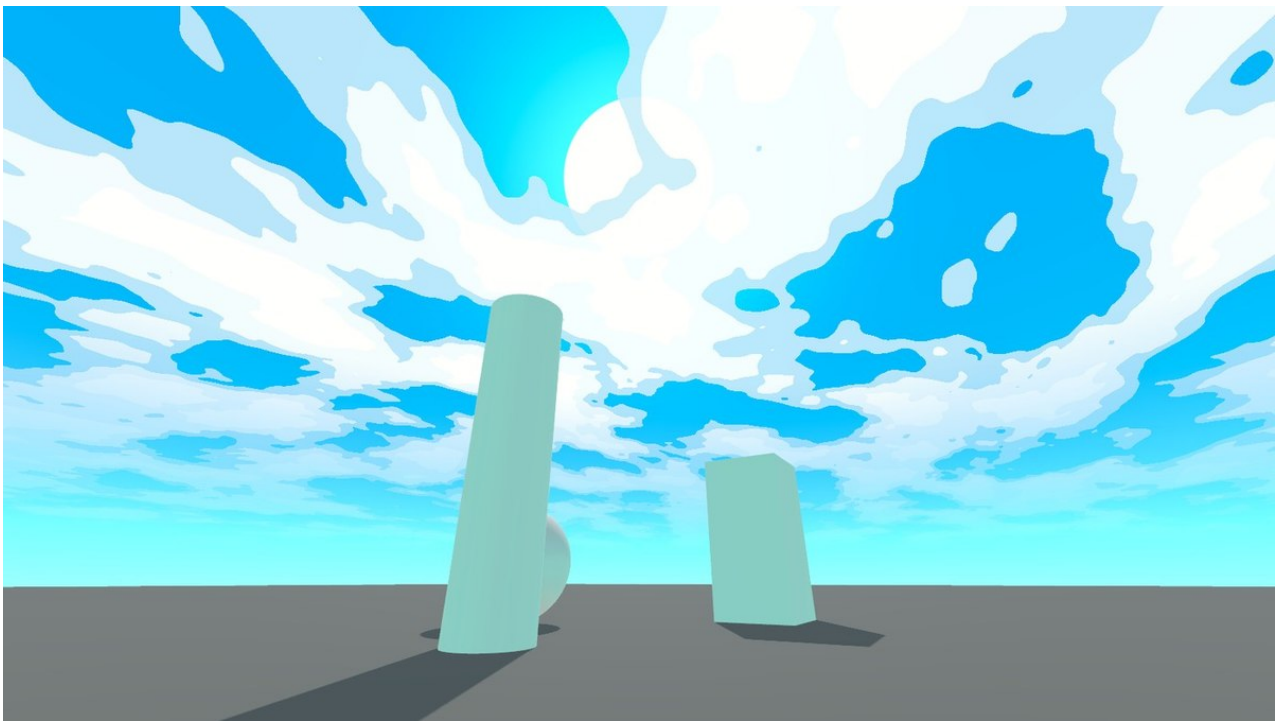
cloud_style 1, Feathered: thin torn wisps drawn out across the sky.

1 - Feathered. Fibrous streaks with sharp ridges, in the manner of high cirrus. Good for a high, thin sky that is not meant to cover the scene.



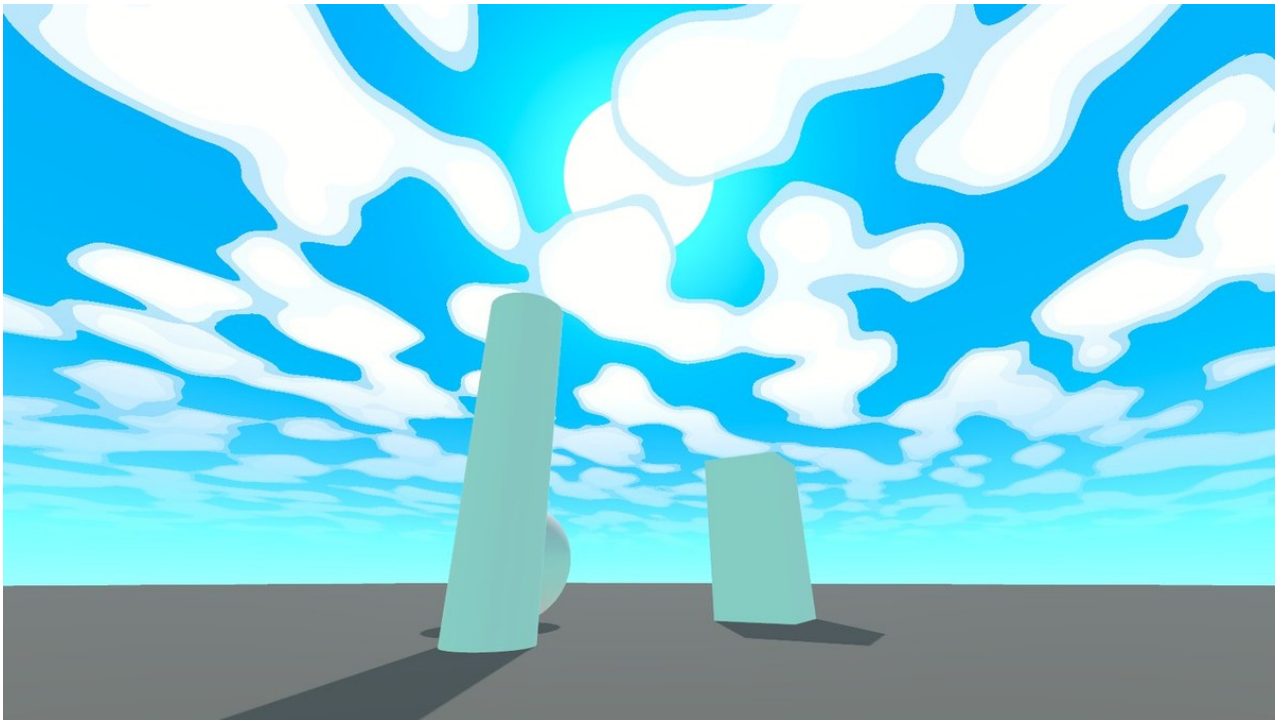
cloud_style 2, Round: separate rounded puffs with a clear gap between neighbours.

2 - Round. Separate rounded puffs, each standing on its own with a visible gap to its neighbour. A drawn look that takes posterization well.



cloud_style 3, Diagonal: the standard pattern rotated 45 degrees into slanted bands.

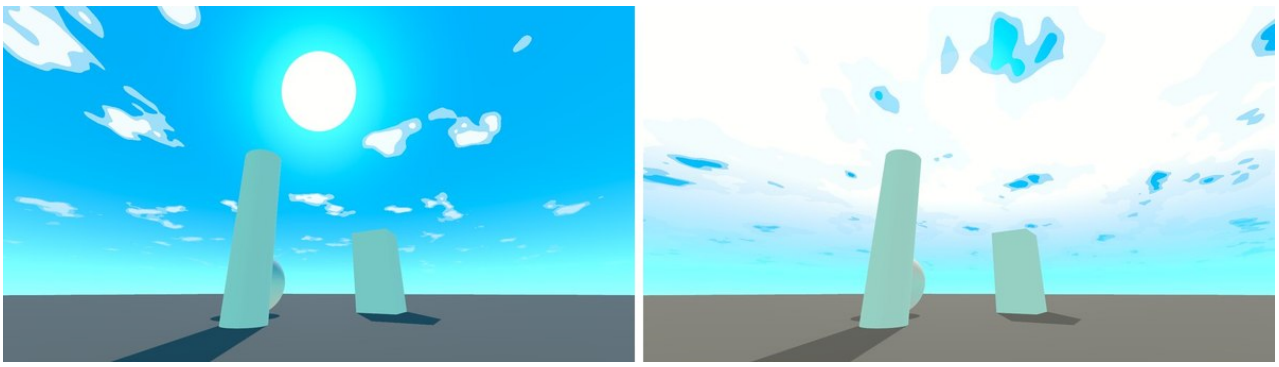
3 - Diagonal. The same pattern as Standard, rotated by 45 degrees. The slanted bands help when horizontal streaks fight the composition of the shot.



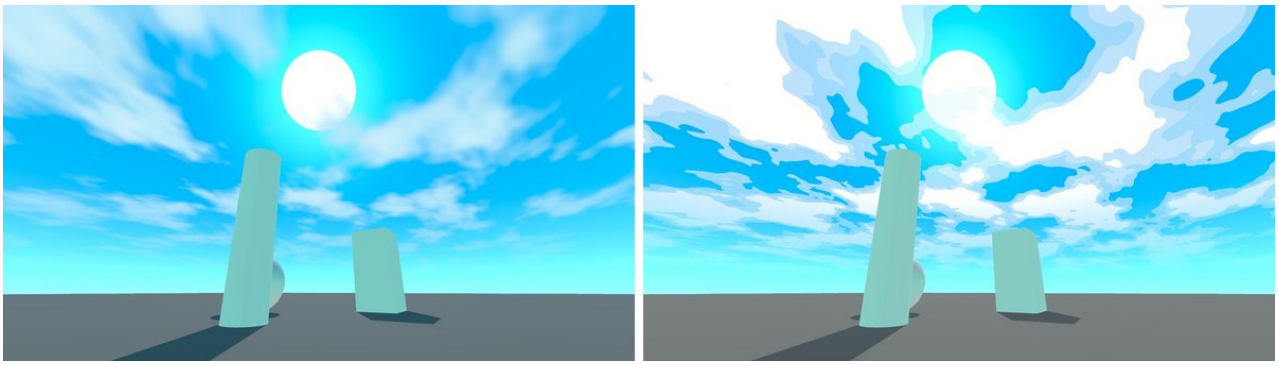
cloud_style 4, Cotton: lobed masses grown together into larger shapes with no seams between puffs.

4 - Cotton. Puffs grown together into larger lobed masses, the way cumulus heaps up. The edge of such a mass wanders, so the shapes never come out as clean circles.

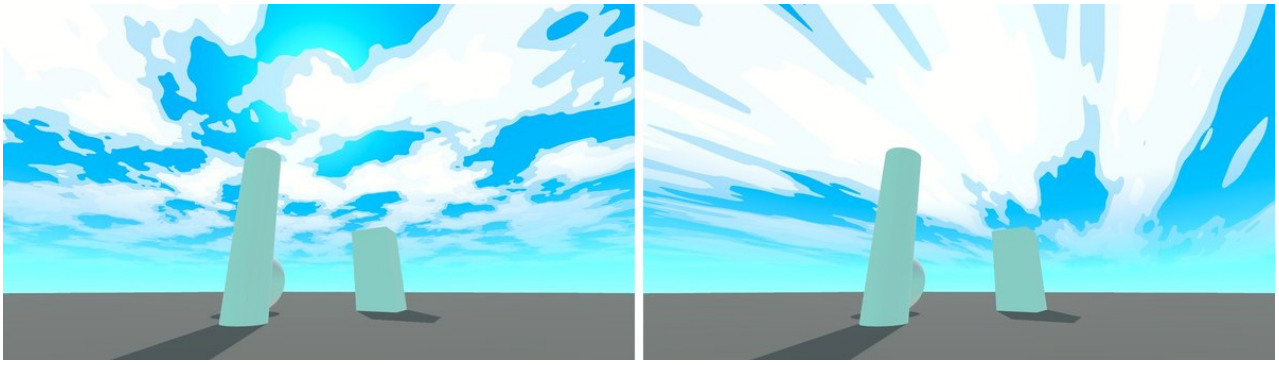
How many clouds, and how hard



Left: cloud_coverage 0.30, single clouds on a clear sky. Right: 0.80, the same pattern grown until it closes most of the sky.



Left: cloud_bands 1, the cloud fades into the sky smoothly. Right: cloud_bands 6, the same cloud in six flat steps of brightness.



Left: cloud_stretch 0.0, the pattern keeps its natural shape. Right: 0.9, the same pattern pulled into long horizontal streaks.

Setting	Range (default)	What it does
clouds_enabled	on / off (**off**)	The master switch for the whole group.
cloud_coverage	0.0 - 1.0 (0.5)	How much of the sky the clouds take up. The cloud edge creeps outward smoothly with the slider, posterization included.
cloud_softness	0.001 - 0.5 (0.15)	Edge softness. Low values give an edge cut with scissors. High values let the cloud dissolve into the sky. The same slider controls how far clouds fade in above the lower limit set by cloud_height.

Setting	Range (default)	What it does
cloud_bands	1.0 - 8.0 (1.0)	**Posterization.** At 1.0 the brightness of the cloud varies smoothly. Above that the cloud gets as many flat steps as the value, for a cel-shaded drawn look. The step count is also the number of shades between cloud_shadow_color and cloud_color.
cloud_density	0.0 - 1.0 (1.0)	Cloud opacity. Below 1.0 the sky shows through the cloud.
cloud_stretch	0.0 - 0.97 (0.3)	Pulls the pattern along one axis into long streaks. It works unevenly on purpose: gentle at the start of the range, dramatic at the end.
cloud_swirl	0.0 - 1.0 (0.0)	Twists the pattern as if the clouds were caught in an eddy. Zero leaves the pattern straight.
cloud_style	0 - 4 (0)	The cloud pattern, described above.
cloud_scale	0.5 - 8.0 (3.0)	Cloud size. Low values give a few huge masses filling the sky. High values give a fine, busy pattern.
cloud_height	0.0 - 0.5 (0.05)	The height below which there are no clouds. Raise it to open a clear strip of sky above the horizon.
cloud_speed	0.0 - 1.0 (0.03)	How fast the clouds drift across the sky. Zero holds them still.
cloud_color	color (white)	The color of the lit part of the cloud.
cloud_shadow_color	color (cool light grey)	The color of the shaded part of the cloud. With cloud_bands above 1, the steps are spread between these two colors.

5. The demo scene



The demo scene right after opening. The panel on the left has a slider for every shader parameter, and the sky changes while you drag.

Open Scenes/SkyboxDemo.tscn and run the scene. The panel on the left holds a slider for every setting from chapter 4, so you can watch what each one does before typing anything into your own project.

Controls:

Key	Action
Left mouse button + move	Look around
Mouse wheel	Zoom in and out
Space	Freeze the cloud drift
H	Hide and show the panel (for a clean screenshot)

Above the sliders there is a **Preset** list with two ready-made looks: Daylight and Warm Dusk. Those are the same two materials you get in the Materials/ folder.

□□ **The demo sliders save nothing to disk.** Close the scene, open it again, and the sky is back where it started; the .tres files in the package stay untouched. When you find a look you like, **write the numbers down** from the panel and set them on your own material in the inspector.

6. What is in the package

Path	What it is
Shaders/EA_Skybox.gdshader	The sky shader. 38 parameters in six inspector groups.
Materials/EA_Skybox_Default.tres	Ready-made material: clear daylight.
Materials/EA_Skybox_WarmDusk.tres	Ready-made material: warm dusk, low sun, posterized clouds.

Path	What it is
Scenes/SkyboxDemo.tscn	Demo scene: sky, sun, ground and three props.
Scenes/UI_ParameterPanel.tscn	The slider panel used in the demo.
Scripts/DemoCamera.gd	Mouse-driven demo camera.
Scripts/SkyPresets.gd	The preset switch in the demo.
Scripts/UI_ParameterPanel.gd	Makes the sliders work; H hides the panel.
Documentation/	This manual as a PDF, a contents list and the licence.

Textures: none. The shader draws the entire sky.

The scripts in Scripts/ serve the demo scene only. Your own scene needs nothing but the shader and a material, with no script running in the background for the sky to be drawn.

7. Performance and compatibility

Mobile renderer. The **Compatibility renderer has not been tested**.

nothing to load from disk.

Round and **Cotton** are the most expensive. When you are aiming at weaker hardware, stay with Standard.

- **Engine:** Godot 4.7 or newer, **Forward+** renderer. The package also runs on the
- **Cost.** A sky with just the gradient, the sun and the haze is cheap, and there is
- **Clouds cost the most**, which is why they have their own switch. Of the five patterns,
- **Cloud drift** (cloud_speed above zero) costs nothing beyond clouds standing still.

8. Licence

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EmacEArt - Maciej. Questions and reports: through the package page on the store.